

Part A

Using the files above fit the CAPM model $r_i - r_f = \alpha_i + \beta_i (r_M - r_f) + \varepsilon$ for each of the 10 stocks, and report whether CAPM “holds” for each (i.e., test $\alpha_i = 0$). Paste prompts, code, output.

AI Code:

```
import pandas as pd
import numpy as np
import statsmodels.api as sm

# --- Load data ---
stocks = pd.read_csv('m_logret_10stocks.txt', sep='\t')
mkt = pd.read_csv('m_sp500ret_3mtcm.txt', sep=r'\t+', skiprows=1, engine='python')
mkt.columns = ['Date', 'sp500', 'tcm3m']

# --- Parse dates to year-month keys so the two different date formats line up ---
stocks['ym'] = pd.to_datetime(stocks['Date']).dt.to_period('M')
mkt['ym'] = pd.to_datetime(mkt['Date'], format='%b-%y').dt.to_period('M')

df = pd.merge(stocks.drop(columns='Date'), mkt.drop(columns='Date'), on='ym', how='inner')
df = df.sort_values('ym').reset_index(drop=True)

# --- Convert annualized 3-month T-bill rate (%) to a monthly risk-free rate ---
df['rf'] = df['tcm3m'] / 100 / 12

tickers = ['AAPL', 'ADBE', 'ADP', 'AMD', 'DELL', 'GTW', 'HP', 'IBM', 'MSFT', 'ORCL']

results = []
for t in tickers:
    y = (df[t] - df['rf']).rename('excess_ret')      #  $r_i - r_f$ 
    x = (df['sp500'] - df['rf']).rename('mkt_excess') #  $r_M - r_f$ 
    X = sm.add_constant(x)
    model = sm.OLS(y, X).fit()
    alpha = model.params['const']
    beta = model.params['mkt_excess']
    t_alpha = model.tvalues['const']
    p_alpha = model.pvalues['const']
    results.append({
        'Stock': t,
        'Alpha': alpha,
        'Alpha_t': t_alpha,
        'Alpha_p': p_alpha,
```

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'Beta': beta,
'Beta_t': model.tvalues['mkt_excess'],
'R2': model.rsquared,
'N': int(model.nobs),
'CAPM_holds(alpha=0 at 5%): 'Yes' if p_alpha > 0.05 else 'No'
})

```

```

res_df = pd.DataFrame(results)
pd.set_option('display.float_format', lambda v: f'{v:0.4f}')
print(f'N observations after merge: {len(df)}')
print(df['ym'].min(), 'to', df['ym'].max())
print()
print(res_df.to_string(index=False))

```

Code Output:

```

N observations after merge: 156
1994-01 to 2006-12

```

Stock	Alpha	Alpha_t	Alpha_p	Beta	Beta_t	R2	N	CAPM_holds(alpha=0 at 5%)
AAPL	0.0038	0.7614	0.4476	1.3846	4.8942	0.1346	156	Yes
ADBE	0.0047	0.9529	0.3422	1.5314	5.5950	0.1689	156	Yes
ADP	0.0008	0.4289	0.6686	0.8477	8.0472	0.2960	156	Yes
AMD	-0.0006	-0.1046	0.9168	2.3238	7.0185	0.2423	156	Yes
DELL	0.0089	2.1637	0.0320	1.6750	7.2865	0.2564	156	No
GTW	-0.0054	-0.9647	0.3362	2.2328	7.0916	0.2462	156	Yes
HP	0.0019	0.5213	0.6029	0.8752	4.2883	0.1067	156	Yes
IBM	0.0026	1.0949	0.2753	1.3479	10.0400	0.3956	156	Yes
MSFT	0.0043	1.4533	0.1482	1.4585	8.8970	0.3395	156	Yes
ORCL	0.0039	0.8809	0.3798	1.5676	6.3071	0.2053	156	Yes

Part B

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$$\beta = \frac{\text{Cov}(r_i - r_f, r_M - r_f)}{\text{Var}(r_M - r_f)} = \frac{0.00027217}{0.00032108} = 0.8476704$$

This is the same as the beta found from the fitted model for ADP in Part A.

Part C

The first thing that is wrong is using incorrect logic for the t-statistic for significance, the t-statistic being 0.4 is low. Typically, for a 95% confidence level, the t-test threshold is at least 1.96, so a 0.4 would mean that the coefficient is highly insignificant and there is a possibility that beta is zero. Second, the AI also has incorrect logic for what a large alpha t-statistic means for CAPM.

If alpha has a large t-statistic, alpha is significantly different from zero, which is evidence against CAPM. The AI has the CAPM null-hypothesis test backwards.

Part D

With a beta of about 0.8477, stock #3 (ADP) is a defensive investment since having a beta less than 1 means that, in theory, it is less volatile than the border market. If CAPM is rejected for a stock, then it means the stock is earning returns that its market risk exposure alone doesn't explain. Economically, that's a sign the single-factor market model is missing something, such as other priced risk factors or the market is inefficient and there's mispricing to exploit. The assumption to worry about most is independence of errors (no autocorrelation), because financial returns often cluster or carry over momentum from month to month, which violates the requirement that each month's market shock is completely random and independent